MUHAMMAD IMRAN

BT301-Introduction-to-Biotechnology — QUIZ NO 1

1. **Deoxyribose** **a pentose** sugar derivative.
2. Bacterial Artificial Chromosomes (BACs) were used **to prepare genomic libraries**.
3. DNA with bound histones in the eukaryotes is called as **chromatin.**
4. Sumerians & Babylonians used anaerobic respiration of yeast to make beer in **6000 bc.**
5. (Genomes of Bacteria Small organisms carry high coding density **(85-90%)**
6. In humans ~ **23,000** Bacterial genomes ~ **1,500 – 2,000** genes.
7. **Highly conserved** genes have changed very little over the time
8. Yeast genome was completely sequenced by **1996**
9. Formed by duplication of an ancestral gene followed by mutation
10. refers to the conventional techniques that have been used for many centuries to produce beer, wine, cheese etc**.** **none of these**
11. Roche acquired Genentech for **$46.8** billion
12. The successive nucleotides of DNA are joined together through **phospho-diester** linkages.
13. **Transfer RNAs** serve as adapter molecules in the process of protein synthesis
14. Use of salts to preserve food specially meat in **400 BC**
15. **In 1928** Alexander Fleming discovers penicillin**, Penicillium Notatum**
16. A revolutionary discovery in the field of biotechnology come in 1973 when **restriction enzyme** were discovered by boyer.
17. 1878 **Albrecht Kossel** isolated non-protein component of nuclein & discovered nucleic acids & isolated 5 bases: A, T, G, C & U.
18. The process of forming mRNA on a DNA template is known as **transcription.**
19. **Watson and Crick postulated** a three dimensional model of DNA structure **in 1953.**
20. Bond exist between cytosine and guanine is **triple bond.**
21. The branch of biotechnology concerned with the discovery and development of innovative drugs and treatments is called **Red Biotechnology**
22. DR. WINSTON saiser was the chairman of **amgen** company
23. **Transfer** RNAs serve as adapter molecules in the process of protein synthesis
24. **The nucleotide** that is present in RNA and absent in DNA is **uracil.**
25. **ATCG** is a building blocks of **DNA**
26. **Urea** is synthesizing in 1828 by **Fredrick Wohler**.
27. **mRNA** is always **single** stranded when it is formed from DNA.
28. Which is the following pair of nucleotides is **purines.** **A ,G**
29. LTRs consist of **250-600** bp.
30. Conjugative plasmid carry set of **tra gene** for Conjugation.
31. A Southern blot is a laboratory method used to detect specific **DNA** .
32. most commonly used enzyme for joining blunt-ended DNA molecules **T4 DNA ligase.**
33. Nick produced in the DNA strand are repaired by **DNA ligase.**
34. Cscl.etbr based centrifugation is used or purification of…….
35. A Southern blot is a laboratory method used to detect **specific** DNA.
36. The **plasmid** commonly exist as cccDNA.
37. The **linker** are blunt ended dna molecules while **adaptor** have cohesive end.
38. In autoradiography we used **X-ray** film to visualize the radioisotope.

39. Probe is singe stranded DNA/RNA which is radiolabeled with **p32** to detect complementary sequence.

40. 1876 Kuhne creates the term “enzymes” and discovered **trypsin.**

41. nuclein discovered in .....(1869).

42. organisms contain single circular DNA .....(bacteria)

43. *coli’s* chromosome consists of approx. **4 million** nucleotide pairs.

44. A multi-subunit protein is also referred to as a **multimer**.

45. RNA act as template **....(mRNA)**

46. genome research help for....(all)DNA forensic ,agriculture

47. we sequence genome.....(**all)identity ,regulation**

48. Schwann proposed cell theory in**.....(1834)(1839)**

49. The repeating units in the nucleic acids are **‘Nucleotides’**.

50. enzyme remove phosphate group(acidic phosphate or alkaline phosphate or alkaline kinase)

51. DNA bands can be visualized by soaking the gel in a solution of **ethidium bromide**

52. E.coli have supercoiled loops....(400bp)

53. It is the type of DNA that move within genomes . **retrotransposons**

54. **Col E1** is a natural occurring plasmid.

55. Human mitochondrial chromosome consists of 37 Genes. Out of which 22 are tRNAs, 2 are rRNAs and **13 are polypeptides**.

56. Pauling and Corey predicted a second type of secondary structure which they called **β-sheets..**

57. Cell lyses by SDS and **proteinase K**

58. covlent bond exit b/w nitrogen base and Penrose sugar.

59. Northern blotting can be used to analyze a sample of RNA from a particular tissue or cell type in order to measure the RNA expression of particular genes

60. PcR is used to amplify Technique to amplify the **DNA.**

61. tyPes of restriction modification **FOUR**.

62. Purines consist of.....**ADENINE GUANINE.**

63. IS10 is found in R plasmids

64. Humans have genome size of **3,000 Mb.**

65. An *E. coli* chromosome is estimated to have about **400 super-coiled** loops.

66. minisatellites have.. **Repeats of 14-500 bp Microsatellites Repeats up to 13 bp.**

67. The minor groove of DNA have..**3.4..Angston.**

68. correct defination of Genetics... Genetics is the study of heredity, or how the characteristics of living organisms are transmitted from one generation to the next generation through DNA.

69. Mainly **three methods** are used for joining DNA in vitro.

70. Bacteria) Small organisms carry high coding density (85-90%).

71. Yeast genome was completelysequenced by 1996

72. LINES • 1-5 kb

73. It is concerned with the discovery and development of innovative drugs and treatments. ? Red biotechnology

74. Nucleotide differ due to change in??? bases

75. The hydrophilicbackbones ofalternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water.

76. mRNA transcribe the template into Protein

77. Capsid protects the Nucleic acid in virus.

78. The bacterial chromosome is Avery long (up to 1mm).

79. Genomics, in contrast, is the stud of the entirety of an organism’s genes –called the genome

80.